COMPUTER ENGINEERING

QUEZON CITY

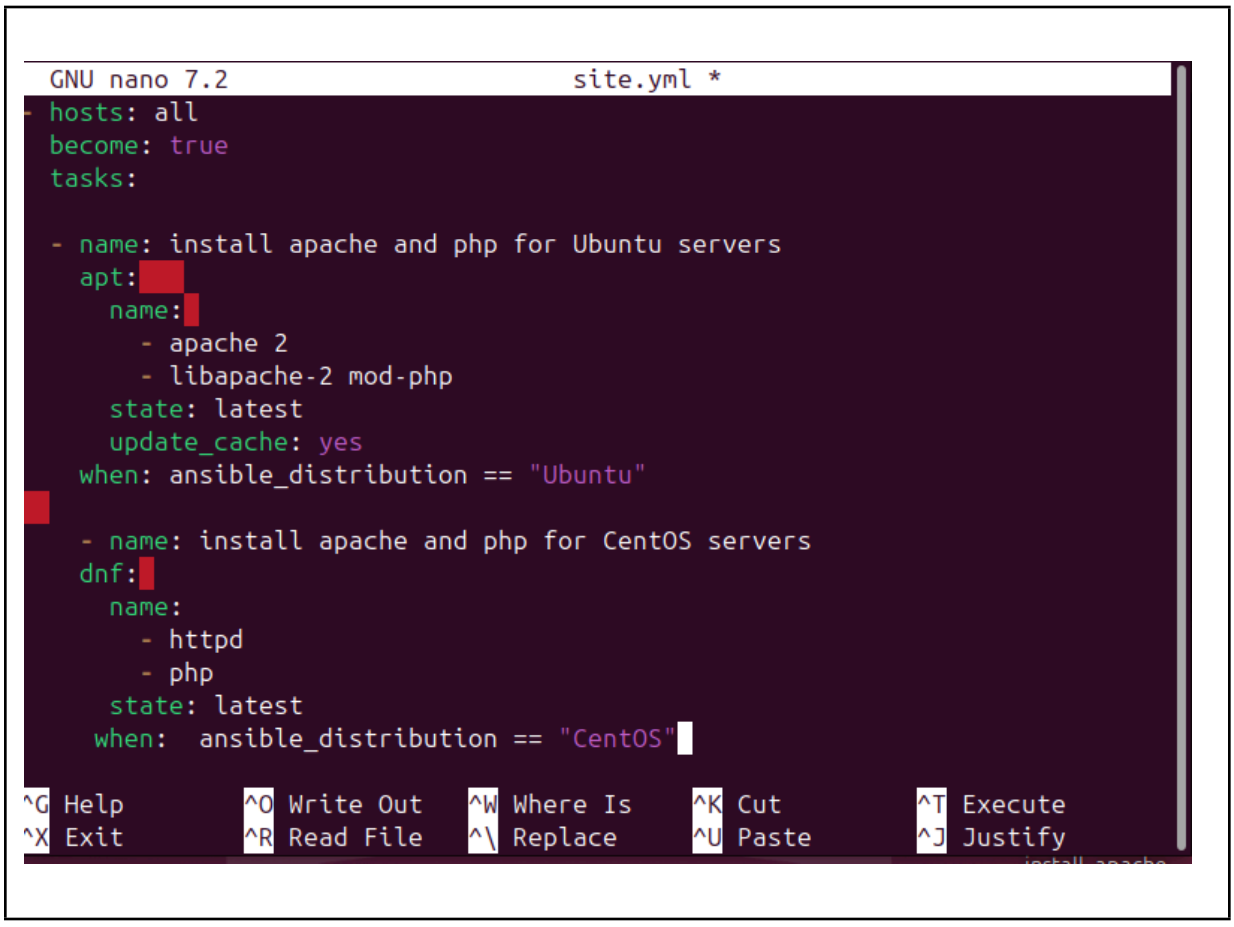
COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	September 4, 2026	**DATE SUBMITTED:**	September 9, 2026
NAME:	Nadine A. Malabanan	**INSTRUCTOR:**	Engr. Jimlord Queado

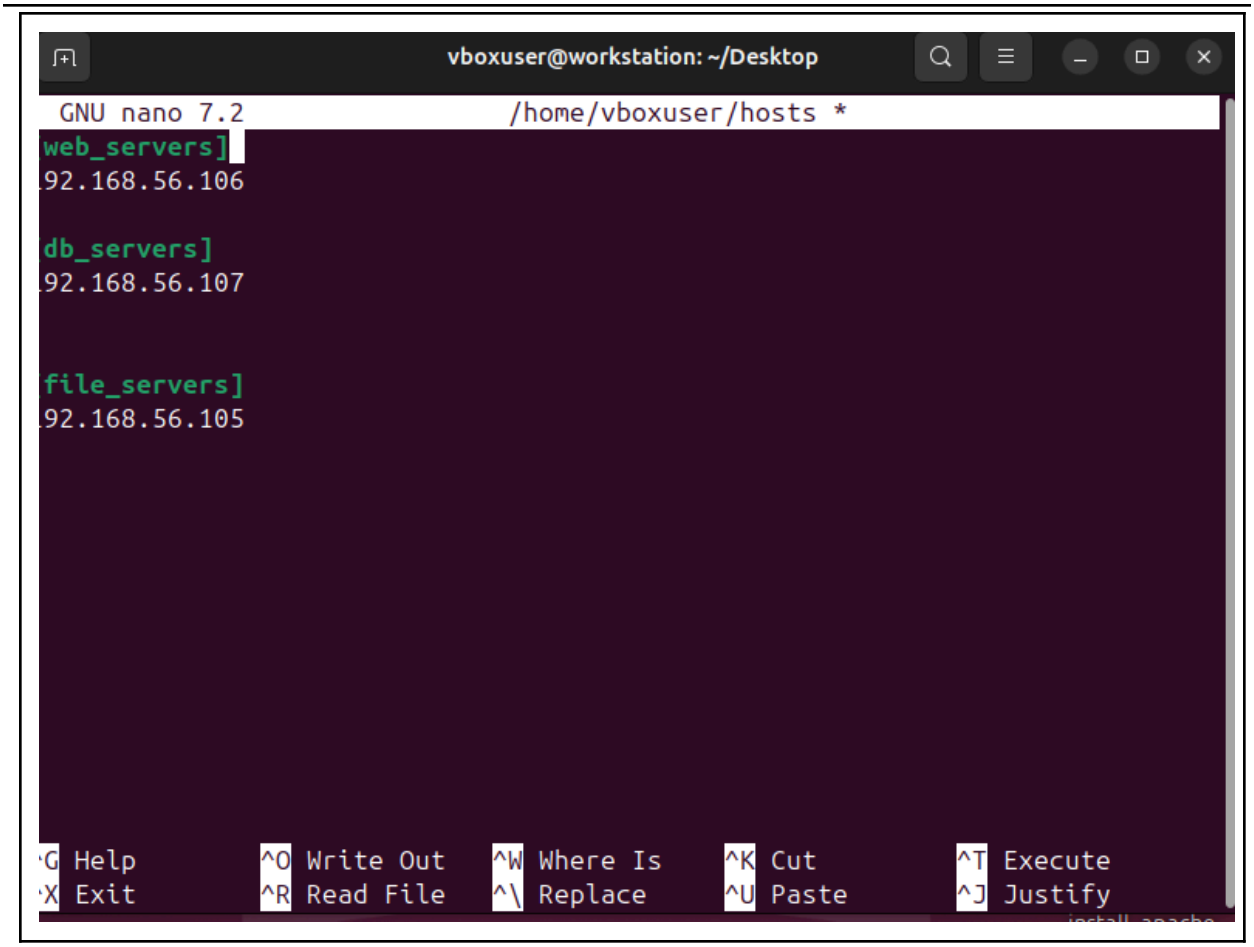
ACTIVITY NO. __6.1_ Targeting Specific Nodes

1. PROCEDURE OUTPUT

What procedures did you perform?

Task 1: Targeting Specific Nodes




vboxuser@workstation: ~/Desktop
GNU nano 7.2 /home/vboxuser/hosts *
web_servers]
92.168.56.106
db_servers]
92.168.56.107
file_servers]
92.168.56.105
G Help
X Exit
^O Write Out
^R Read File
^W Where Is
^\ Replace
^K Cut
^U Paste
^T Execute
^J Justify

```
vboxuser@workstation:~/Desktop$ ansible all -i ~/hosts --list-hosts
  hosts (3):
    192.168.56.106
    192.168.56.107
    192.168.56.105
vboxuser@workstation:~/Desktop$ cat site.yml
---

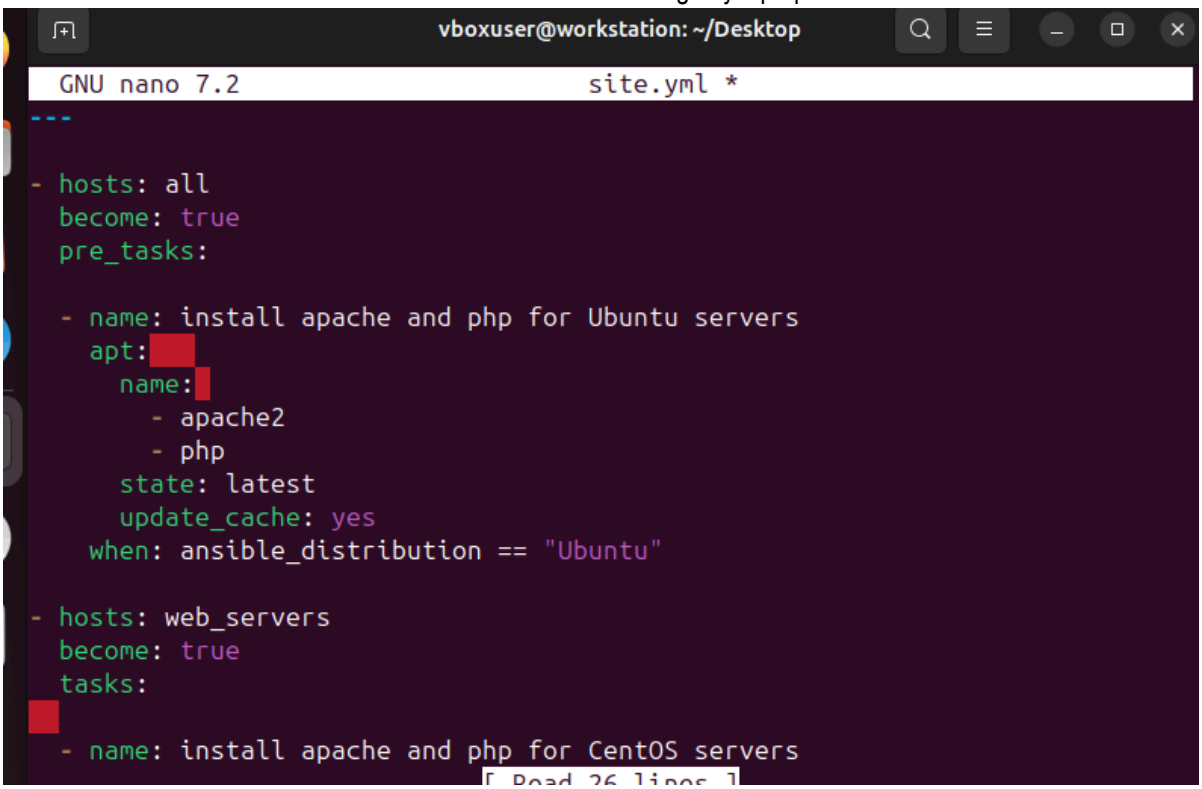
- hosts: all
  become: true
  tasks:

  - name: install apache and php for Ubuntu servers
    apt:
      name:
        - apache 2
        - libapache-2 mod-php
      state: latest
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

  - name: install apache and php for CentOS servers
    dnf:
```

```
vboxuser@workstation:~/Desktop$ ansible all -i ~/hosts -m ping --ask-pass
SSH password:
192.168.56.106 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
192.168.56.105 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
192.168.56.107 | UNREACHABLE! => {
    "changed": false,
    "msg": "Failed to connect to the host via ssh: ssh: connect to host 192.1
68.56.107 port 22: No route to host",
    "unreachable": true
}
vboxuser@workstation:~/Desktop$
```

The CentOS was unreachable because the VMs was not running. My laptop cannot handle four VMs at once.




```
changed: [192.168.56.105]
changed: [192.168.56.106]

PLAY [web_servers] *************************************************************
***

TASK [Gathering Facts] *********************************************************
***
ok: [192.168.56.106]

TASK [install apache and php for CentOS servers] *******************************
***
skipping: [192.168.56.106]

PLAY RECAP *********************************************************************
***
192.168.56.105             : ok=2    changed=1    unreachable=0    failed=0
  skipped=0    rescued=0    ignored=0
192.168.56.106             : ok=3    changed=1    unreachable=0    failed=0
  skipped=1    rescued=0    ignored=0
192.168.56.107             : ok=0    changed=0    unreachable=1    failed=0
  skipped=0    rescued=0    ignored=0
```

4.

```
  GNU nano 7.2                          site.yml *
  tasks:

  - name: install mariadb package (CentOS)
    yum:
      name: mariadb-server
      state: latest
    when: ansible_distribution == "CentOS"

  - name: "Mariadb- Restarting/Enabling"
    service:
       name: mariadb
       state: restarted
       enabled: true

  - name: install apache and php for CentOS servers
    apt:
      name: mariadb-server
      state: latest
    when:  ansible_distribution == "Ubuntu"

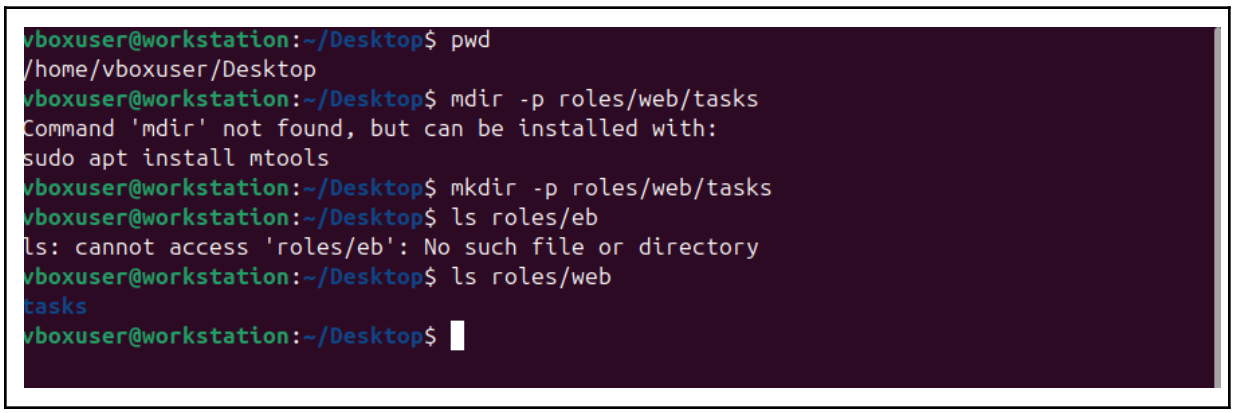

^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify
```

```
  GNU nano 7.2                          inventory
[web_servers]
192.168.56.106 ansible_user=vboxuser
[db_servers]
192.168.56.105 ansible_user=vboxuser
[file_servers]
```

2. ANSWE RS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and
include your answers after.

Supplementary Activity:
1. Creating a Role for Web Servers
2. Creating a Role for Web Servers
3. Creating a Role for Database Servers
4. Combining Multiple Roles in a Playbook
To orchestrate multiple roles in a single playbook for different server groups.
Add tags to each role to allow selective execution, e.g., tags: web for webserver, tags: db for Database.



```
 GNU nano 7.2                    roles/web/tasks/main.yml *
---
- name: Install Apache
  apt:
    name: apache2
    state: latest
    update_cache: yes

- name: Start Apache
  service:
     name: apache2
     state: started
     enabled: true
```

```
roles/web/tasks:
vboxuser@workstation:~/Desktop$ nano roles/eb/tasks/main.yml
vboxuser@workstation:~/Desktop$ nano roles/web/tasks/main.yml
vboxuser@workstation:~/Desktop$ cat roles/web/tasks/main.yml
---
- name: Install Apache
  apt:
    name: apache2
    state: latest
    update_cache: yes

- name: Start Apache
  service:
     name: apache2
     state: started
     enabled: true
vboxuser@workstation:~/Desktop$
```

```
 GNU nano 7.2                    roles/db/tasks/main.yml *
---
- name: Install MariaDB
  apt:
    name: mariadb-server
    state: latest
    update_cache: yes

- name: Start MariaDB
  service:
    name: mariadb
    state: started
    enabled: yes
```

```
vboxuser@workstation:~/Desktop$ ls -R
.:
hosts               install_apache.yml.save  roles     site.yml
install_apache.yml  inventory                site,yml

./roles:
db  web

./roles/db:
tasks

./roles/db/tasks:
main.yml

./roles/web:
tasks

./roles/web/tasks:
main.yml
```

```
 GNU nano 7.2                          site.yml *
- hosts: web_servers
  become: true
  roles:
    - role: web
      tags:
        - web

  - hosts: db_servers
    become: true
    roles:
      - role: db
        tags:
          - db
```

```
vboxuser@workstation:~/Desktop$ cat site.yml
- hosts: web_servers
  become: true
  roles:
    - role: web
      tags:
        - web

  - hosts: db_servers
    become: true
    roles:
      - role: db
        tags:
          - db
```

```
vboxuser@workstation:~/Desktop$ nano site.yml
vboxuser@workstation:~/Desktop$ ansible-playbook -i inventory site.yml --syntax-check

playbook: site.yml
vboxuser@workstation:~/Desktop$
```

```
vboxuser@workstation:~/Desktop$ ansible-playbook -i inventory site.yml --ask-pass --ask-
become-pass
SSH password:
BECOME password[defaults to SSH password]:

PLAY [web_servers] ************************************************************************

TASK [Gathering Facts] ********************************************************************
ok: [192.168.56.106]

TASK [web : Install Apache] ***************************************************************
```

Reflections:
Answer the following:
1. What is the importance of putting our remote servers into groups?
Putting remote servers into groups helps us organize them based on their purpose, such as web, database, and file servers. It also allows us to target only the servers that need a specific configuration or task.

2. What is the importance of tags in playbooks?
Tags are important because they allow us to run only specific tasks in a playbook instead of running everything. This makes the playbook more flexible and saves time when managing servers.

3. Why do you think some services need to be managed automatically in playbooks?
Some services need to be managed automatically because installing a package does not always mean that its service will start automatically. Using Ansible, we can make sure that the service is started and enabled whenever the playbook is executed.

Conclusions/Learnings:

In this activity, I learned how to group remote servers based on their roles and target specific servers using Ansible. I also learned how tags can be used to run only selected tasks and how services such as MariaDB and Apache can be managed through playbooks. This activity helped me understand how Ansible can make server management more organized, easier, and less repetitive.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **N/A**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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